

Errata for *A 2 million-person, campaign-wide field experiment shows how digital advertising affects voter turnout*

Seven corrections to *Nature Human Behaviour* article 10.1038/s41562-022-01487-4. None of them changes a conclusion of the article. Every corrected value below is computed at render time from a full reproduction of the published analysis, held in the repository `active-maintenance-aec/aggarwal_etal_2023`. Bold marks the tokens that changed; everything else in each corrected sentence is the published wording.

1. The month of heaviest persuasion spending

In fact, we observe significant early spending (Acronym spent more on persuasion in **July** than in any other month, see main Figure ??), and we find much stronger effects on early voting than day of voting (see Figure 3 in the main text).

Corrected: In fact, we observe significant early spending (Acronym spent more on persuasion in **February** than in any other month, see **Supplementary Figure 2**), and we find much stronger effects on early voting than day of voting (see Figure 3 in the main text).

The figure the sentence points at is Supplementary Figure 2, whose two panels split Acronym's monthly lower-bound spending on persuasion advertisements by format and by keyword. Summed over either panel's series, the largest month is February 2020 at US\$577,000, against US\$566,600 in July. The cross-reference did not resolve when the document was typeset. The sentence's argument, that Acronym spent heavily earlier in the campaign than other left-leaning advertisers did, is unaffected: February is earlier than July.

2. The number of pre-treatment covariates tested for balance

As shown in Fig. 5, the random assignment procedure generated treatment and control groups that were balanced on pre-treatment characteristics in this subset. Out of **21** covariates

Corrected: As shown in Fig. 5, the random assignment procedure generated treatment and control groups that were balanced on pre-treatment characteristics in this subset. Out of **22** covariates

Figure 5 plots twenty-two covariates and Supplementary Table 3 prints twenty-two rows. The count of two significant imbalances is right, and so is everything the paragraph goes on to say about them.

3. The size of the treatment audience

The full cost of the advertising campaign was US\$8.9 million, spread out over a treatment audience of **1,993,216 million** (3,322,027 programme-eligible voters $\times$ our 60% match rate), amounting to US\$4.46 of advertising expenditure per voter.

Corrected: The full cost of the advertising campaign was US\$8.9 million, spread out over a treatment audience of **1,993,216** (3,322,027 programme-eligible voters $\times$ our 60% match rate), amounting to US\$4.46 of advertising expenditure per voter.

The parenthetical states the arithmetic: 3,322,027 times a 60 per cent match rate is 1,993,216 people. The word “million” makes the figure a million times too large.

4. The degrees of freedom of the balance F test

An F-test comparing a regression of treatment status on covariates with strata fixed effects versus a restricted model predicting treatment status from strata fixed effects was non-significant ($F_{1999264, -18} = 1.0266$; $P = 0.425$).

Corrected: An F-test comparing a regression of treatment status on covariates with strata fixed effects versus a restricted model predicting treatment status from strata fixed effects was non-significant ($F_{18, 1999246} = 1.0266$; $P = 0.425$).

An F statistic carries its numerator degrees of freedom first. The unrestricted model has 18 more parameters than the restricted one and leaves 1,999,246 residual degrees of freedom. The published subscripts are the second row of the Wald test’s printed output, where the first figure is that model’s residual degrees of freedom and the second is the difference in parameters, printed as a negative number. The statistic and the p-value are right.

5. Acronym’s share of federal advertising spending on Facebook

According to this data, there was a total of approximately \$977,761,865 spent on federal races on Facebook from 2/1/2020 to 11/3/2020, \$6,603,488 (**0.67%**) of which was from Acronym.

Corrected: According to this data, there was a total of approximately \$977,761,865 spent on federal races on Facebook from 2/1/2020 to 11/3/2020, \$6,603,488 (**0.68%**) of which was from Acronym.

A rounding artifact rather than a substantive error: the quotient of the sentence’s own two figures is 0.6754 per cent, which rounds up. The share is kept at two decimal places because at one decimal the published and corrected figures are the same.

6. The definition of the black variable in the randomisation code

```
, case when vb_voterbase_race = 'African-American' or (vb_voterbase_race = 'Uncoded' and civis_race
```

Corrected: , case when vb_voterbase_race = 'African-American' or (vb_voterbase_race = 'Uncoded' and civis_race = 'AFAM') then 'black' else null end as black

The line is drawn to x 610.6 on a page 612 points wide and the text layer ends with `civis_race`. The complete line is in the authors' own `Acronym/notes/gh_randomization_sql.mssql`, line 17, which agrees with every other line of the published listing except for the capitalisation of the schema name. `black` is one of the two race categories the Methods report as no longer available, and the code as printed cannot be run.

7. The definition of the hispanic variable in the randomisation code

```
, case when vb_voterbase_race = 'Hispanic' or (vb_voterbase_race = 'Uncoded' and civis_race = 'HISPA
```

Corrected: , case when vb_voterbase_race = 'Hispanic' or (vb_voterbase_race = 'Uncoded' and civis_race = 'HISPANIC') then 'hispanic' else null end as hispanic

The line is drawn to x 615.8 on a page 612 points wide, so it ends beyond the page edge and mid-token, at `'HISPA`. The complete line is in the authors' own `Acronym/notes/gh_randomization_sql.mssql`, line 18. `hispanic` is the other race category the Methods report as no longer available.